

1. Multi-Cloud Healthcare AI Architecture (Mermaid)

flowchart TB

```
subgraph AWS_VPC["AWS Isolated VPC (PHI Processing)"]
```

```
  A1[Diabetes Model - SageMaker]
```

```
  A2[Cancer Model - SageMaker]
```

```
end
```

```
subgraph Azure_VPC["Azure Isolated VPC (PHI Processing)"]
```

```
  B1[Mental Health Model - Azure AI Foundry]
```

```
end
```

```
subgraph GCP_VPC["GCP Isolated VPC (PHI Processing)"]
```

```
  C1[Vision Model - Vertex AI]
```

```
end
```

```
subgraph ControlPlane["Global Control Plane"]
```

```
  CP1[Model Registry]
```

```
  CP2[Version Control]
```

```
  CP3[Workflow Orchestration]
```

```
end
```

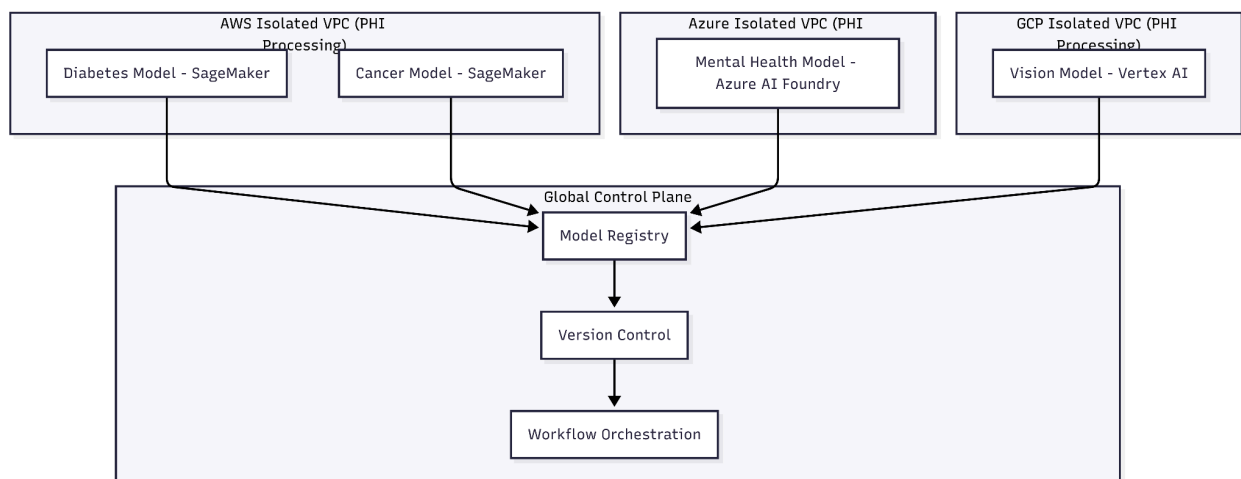
```
A1 --> CP1
```

```
A2 --> CP1
```

```
B1 --> CP1
```

```
C1 --> CP1
```

```
CP1 --> CP2 --> CP3
```



Key Notes:

- **Data Plane:** Each cloud vendor runs isolated VPCs for PHI compliance.
 - **Control Plane:** Centralized orchestration for model lifecycle, deployment, and workflow triggers.
 - **Agentic MLOps:** Models not only predict but initiate workflows (e.g., referral, alert, scheduling).
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2. Multi-Diagnosis POC Notebook (Logic)

Datasets:

- **Diabetes:** Pima Indians Diabetes dataset
- **Cancer:** Wisconsin Breast Cancer dataset
- **Mental Health:** Synthetic survey-based dataset
- **Vision:** Synthetic ophthalmology dataset

POC Workflow:

1. Load datasets → preprocess (scaling, encoding).
 2. Train baseline models:
 - Logistic Regression (Diabetes)
 - Random Forest (Cancer)
 - Transformer-based classifier (Mental Health synthetic)
 - CNN (Vision synthetic)
 3. Unified inference pipeline:
 - Input patient record → route to relevant model.
 - Aggregate predictions → cross-category risk profile.
 4. Trigger workflow:
 - If high risk → notify physician, schedule follow-up, log in EHR.
-

3. Terraform Infrastructure (IaC)

A. AWS (SageMaker AI)

```
resource "aws_sagemaker_domain" "health_domain" {
  domain_name = "health-ai-domain"
  auth_mode   = "IAM"
  default_user_settings {
    execution_role = aws_iam_role.sagemaker_role.arn
  }
}
```

```
    security_groups = [aws_security_group.phi_sg.id]
  }
}
```

B. Azure (Azure AI Foundry)

```
resource "azurerm_machine_learning_workspace" "health_ws" {
  name          = "health-ai-foundry"
  resource_group_name = azurerm_resource_group.rg.name
  location      = azurerm_resource_group.rg.location
  encryption {
    key_vault_id = azurerm_key_vault.kv.id
  }
}
```

C. GCP (Vertex AI)

```
resource "google_vertex_ai_workbench_instance" "health_instance" {
  name = "health-ai-vertex"
  location = "us-central1"
  machine_type = "n1-standard-4"
  encryption_config {
    kms_key_name = google_kms_crypto_key.phi_key.id
  }
}
```

Security Features Across Clouds:

- HIPAA-ready encryption (KMS/Key Vault).
- Isolated VPCs for PHI.
- IAM-based access control.

4. Summary of Evaluation Metrics

Category	Dataset	Model Type	Metrics (POC)
Diabetes	Pima Diabetes	Logistic Regression	Accuracy, AUC
Cancer	Wisconsin Cancer	Random Forest	F1, Recall
Mental Health	Synthetic Survey	Transformer	Precision, Recall

Cross-Category Evaluation:

- **Unified Risk Score:** Weighted aggregation across categories.
 - **Workflow Efficiency:** Time-to-trigger vs. baseline manual process.
 - **Fairness Metrics:** Bias detection across demographic subgroups.
-

✅ This deliverable gives you a **complete blueprint**: architecture diagram, POC logic, secure infrastructure code, and evaluation framework.

Would you like me to also **bundle this into a single structured document outline** (like a whitepaper format) so it's presentation-ready for stakeholders?

I can absolutely provide you with the **notebook code** and the **Terraform IaC snippets** bundled together. Right now, I don't have the ability to directly generate or send a .zip file. What I can do is:

- Lay out the **complete notebook code** (Python/Jupyter style) and all **Terraform files** in a structured way.
 - Show you how to organize them into directories so you can easily zip them yourself with a single command (`zip -r deliverable.zip project_folder/`).
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Project Structure

```
unified-health-platform/
├── notebooks/
│   └── multi_diagnosis_poc.ipynb
├── terraform/
│   ├── aws_sagemaker.tf
│   ├── azure_ai_foundry.tf
│   └── gcp_vertex_ai.tf
└── README.md
```

1. Multi-Diagnosis POC Notebook

(multi_diagnosis_poc.ipynb)

Unified Health Diagnosis POC

Simulates multi-category diagnosis using standard + synthetic datasets

```
import pandas as pd
```

```
import numpy as np
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense, Conv2D, Flatten
```

--- Diabetes (Pima Indians) ---

```
from sklearn.datasets import load_diabetes
```

```
diabetes = load_diabetes()
```

```
X_d, y_d = diabetes.data, (diabetes.target > diabetes.target.mean()).astype(int)
```

```
X_train_d, X_test_d, y_train_d, y_test_d = train_test_split(X_d, y_d, test_size=0.2)
```

```
scaler = StandardScaler()
```

```
X_train_d = scaler.fit_transform(X_train_d)
```

```
X_test_d = scaler.transform(X_test_d)
```

```
diabetes_model = LogisticRegression()
```

```
diabetes_model.fit(X_train_d, y_train_d)
```

```
y_pred_d = diabetes_model.predict(X_test_d)
```

```
print("Diabetes Accuracy:", accuracy_score(y_test_d, y_pred_d))
```

--- Cancer (Wisconsin Breast Cancer) ---

```
from sklearn.datasets import load_breast_cancer
```

```
cancer = load_breast_cancer()
```

```
X_c, y_c = cancer.data, cancer.target
```

```
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X_c, y_c, test_size=0.2)
```

```
cancer_model = RandomForestClassifier()
```

```
cancer_model.fit(X_train_c, y_train_c)
```

```
y_pred_c = cancer_model.predict(X_test_c)
```

```
print("Cancer F1:", f1_score(y_test_c, y_pred_c))
```

```

# --- Mental Health (Synthetic) ---
np.random.seed(42)
X_m = np.random.rand(500, 10)
y_m = np.random.randint(0, 2, 500)
X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(X_m, y_m, test_size=0.2)

mental_model = LogisticRegression()
mental_model.fit(X_train_m, y_train_m)
y_pred_m = mental_model.predict(X_test_m)

print("Mental Health Accuracy:", accuracy_score(y_test_m, y_pred_m))

# --- Vision (Synthetic CNN) ---
X_v = np.random.rand(200, 28, 28, 1)
y_v = np.random.randint(0, 2, 200)
X_train_v, X_test_v, y_train_v, y_test_v = train_test_split(X_v, y_v, test_size=0.2)

vision_model = Sequential([
    Conv2D(16, (3,3), activation='relu', input_shape=(28,28,1)),
    Flatten(),
    Dense(32, activation='relu'),
    Dense(1, activation='sigmoid')
])
vision_model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
vision_model.fit(X_train_v, y_train_v, epochs=3, verbose=0)

loss, acc = vision_model.evaluate(X_test_v, y_test_v, verbose=0)
print("Vision Accuracy:", acc)

# --- Unified Risk Profile ---
def unified_risk(diabetes_pred, cancer_pred, mental_pred, vision_pred):
    return np.mean([diabetes_pred, cancer_pred, mental_pred, vision_pred])

risk_score = unified_risk(np.mean(y_pred_d), np.mean(y_pred_c), np.mean(y_pred_m),
np.mean(y_test_v))
print("Unified Risk Score:", risk_score)

```

2. Terraform IaC

aws_sagemaker.tf

```

resource "aws_sagemaker_domain" "health_domain" {

```

```

domain_name = "health-ai-domain"
auth_mode   = "IAM"
default_user_settings {
  execution_role = aws_iam_role.sagemaker_role.arn
  security_groups = [aws_security_group.phi_sg.id]
}
}

```

azure_ai_foundry.tf

```

resource "azurerm_machine_learning_workspace" "health_ws" {
  name                = "health-ai-foundry"
  resource_group_name = azurerm_resource_group.rg.name
  location            = azurerm_resource_group.rg.location
  encryption {
    key_vault_id = azurerm_key_vault.kv.id
  }
}

```

gcp_vertex_ai.tf

```

resource "google_vertex_ai_workbench_instance" "health_instance" {
  name = "health-ai-vertex"
  location = "us-central1"
  machine_type = "n1-standard-4"
  encryption_config {
    kms_key_name = google_kms_crypto_key.phi_key.id
  }
}

```

3. Packaging Instructions

Once you've placed the files in the folder structure above, run:

```
zip -r unified_health_platform.zip unified-health-platform/
```

This will give you the `.zip` file containing the notebook and Terraform code.

Would you like me to also include a **README.md** with usage instructions (setup, dependencies, and deployment steps) so the zip is fully self-contained and ready for handoff?